

ASSIGNMENT -2

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Main Menu

```
(venv) E:\AI-Powered Customer Support Automation System>python main.py
-> Custom SQLite database initialized at memory.db (table: customer_interactions)

=====
      ABC Technologies - AI-Powered Customer Support Automation System
=====
      Built with: LangGraph, Llama3.2 (Ollama), & SQLite Checkpointing
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Select an option:
[1] Run Automated Demonstration (5 Sample Queries)
[2] Start Interactive Chat Session
[3] View SQLite Database Log Table
[4] Exit

Enter choice (1-4): 1

=== Starting Automated Demonstration ===
All queries will run on the same Thread (demo_thread_001) to demonstrate memory retention.
```

Query 1

DEMONSTRATION QUERY 1/5

--- Processing Query ---

Customer: David

Query: "What are the pricing plans available for your software?"

--- [Node] Classifying Intent ---

-> Classified Intent: Sales (Raw LLM output: 'Sales')

--- [Node] Retrieving RAG Context ---

-> Retrieved context from knowledge base.

--- [Node] Sales Agent ---

--- [Node] Supervisor Agent (AI Validation) ---

-> Supervisor validated and improved the draft response.

--- [Node] Responding to Customer ---

-> Logged interaction to SQLite memory (Thread: demo_thread_001, Dept: Sales)

=== Final Response to Customer ===

Routed Department: Sales

Response: We offer three subscription plans tailored to different business sizes:

* **Basic Plan**: This plan is ideal for small teams and startups, priced at \$29 per user/month (billed monthly) or \$299 per user/year (billed annually - saves ~15%). It includes access to core business management modules, 10 GB secure cloud storage, and standard email support.

* **Professional Plan** (Most Popular): This plan is suitable for growing businesses and mid-sized enterprises, priced at \$79 per user/month (billed monthly) or \$799 per user/year (billed annually - saves ~15%). It includes all Basic Plan features, advanced CRM integrations, project management boards, automated invoicing, and expenses, as well as 100 GB secure cloud storage. Priority email and live chat support are also included.

* **Enterprise Plan**: This plan is designed for large organizations requiring advanced security, custom integrations, and dedicated support. Pricing starts at \$199 per user/month (billed annually) and is customized based on the organization's specific needs. The Enterprise Plan includes all Professional Plan features, unlimited cloud storage, single sign-on (SSO), advanced audit logs, API access, 24/7 dedicated support with a dedicated account manager, and a 99.9% uptime guarantee with credit eligibility.

Please note that the Enterprise Plan is custom-priced based on your organization's requirements, so we recommend contacting our Sales team for a quote tailored to your needs.

Note: No human approval is required for this request.

=====

Press Enter to proceed to the next query...

Query 2

DEMONSTRATION QUERY 2/5

--- Processing Query ---

Customer: David

Query: "I forgot my account password."

--- [Node] Classifying Intent ---

-> Classified Intent: Account (Raw LLM output: 'Account')

--- [Node] Retrieving RAG Context ---

-> Retrieved context from knowledge base.

--- [Node] Account Support Agent ---

--- [Node] Supervisor Agent (AI Validation) ---

-> Supervisor validated and improved the draft response.

--- [Node] Responding to Customer ---

-> Logged interaction to SQLite memory (Thread: demo_thread_001, Dept: Account)

=== Final Response to Customer ===

Routed Department: Account

Response: To reset your password, please follow these steps:

1. Open the login page.
2. Click on "Forgot Password".
3. Enter your registered email address associated with your account.
4. We will send you a password reset link to that email address.

Please note that this link is only valid for 1 hour from when it's sent. If you don't receive the email within 5 minutes, please check your Spam/Junk folder.

Once you click on the link and enter a new password, you'll be able to log in to your account.

Note: No human approval is required for this request.

=====

Press Enter to proceed to the next query...

Query 3

DEMONSTRATION QUERY 3/5

--- Processing Query ---

Customer: David

Query: "My application crashes whenever I upload a file."

--- [Node] Classifying Intent ---

-> Classified Intent: Technical (Raw LLM output: 'Technical')

--- [Node] Retrieving RAG Context ---

-> Retrieved context from knowledge base.

--- [Node] Technical Support Agent ---

--- [Node] Supervisor Agent (AI Validation) ---

-> Supervisor validated and improved the draft response.

--- [Node] Responding to Customer ---

-> Logged interaction to SQLite memory (Thread: demo_thread_001, Dept: Technical)

=== Final Response to Customer ===

Routed Department: Technical

Response: I'd be happy to help you troubleshoot the issue with your application crashing during file uploads.

To start, let's check if the file size exceeds the maximum allowed limit of 50 MB. Can you please verify that your file is within this limit? If it's larger than 50 MB, we can discuss options for compressing or splitting the file to make it compatible for upload.

Additionally, I'd like to confirm that the file format is supported by our application. We only support `.csv`, `.xlsx`, `.pdf`, `.json`, `.jpg`, and `.png` formats. If your file has a different extension, we may need to convert it to one of these formats before uploading.

Lastly, have you checked if there are any network constraints or firewall settings that might be blocking WebSocket connections? These are used for real-time file upload progress updates.

If none of these potential causes seem to be the issue, I'd like to help you investigate further. Can you please provide more details about your environment and application configuration?

Also, just a general question: Are you using our web interface or a custom-built application with our API?

Note: No human approval is required for this request.

=====

Press Enter to proceed to the next query...

Query 4

DEMONSTRATION QUERY 4/5

--- Processing Query ---

Customer: David

Query: "I need a refund for my annual subscription."

--- [Node] Classifying Intent ---

-> Classified Intent: Billing (Raw LLM output: 'Billing')

-> Classified Intent: Billing (Raw LLM output: 'Billing')

--- [Node] Retrieving RAG Context ---

-> Retrieved context from knowledge base.

--- [Node] Billing Support Agent ---

--- [Node] Supervisor Agent (AI Validation) ---

-> Supervisor validated and improved the draft response.

[PAUSED] HIGH-RISK REQUEST DETECTED!

Department: Billing

Risk Category: REFUND

--- Proposed Response (AI Supervisor Polished) ---

To assist you with your refund request for your annual subscription, I'd like to clarify our standard policy regarding refunds.

As stated in our Refund Policy, all refund requests are subject to verification. Since you have an annual subscription, you're eligible for a partial refund if cancelled and requested within the first 30 days of the subscription period. The refund amount will be calculated by deducting the standard monthly rate for the active month(s).

Please note that I've flagged your request as high-risk due to our Escalation Policy, which requires all refund requests to be reviewed by a supervisor before any response or action is executed. This means that your request will need to be escalated to our Human Supervisor for final approval.

Would you like me to proceed with escalating your request?

Supervisor Action Required:

[1] Approve response as-is

[2] Reject response (provide feedback to customer)

[3] Modify response text manually

Enter choice (1, 2, or 3, Default: 1): 2

Supervisor Action Required:

[1] Approve response as-is

[2] Reject response (provide feedback to customer)

[3] Modify response text manually

Enter choice (1, 2, or 3, Default: 1): 2

Enter rejection reason: High Risk

-> Applying Supervisor Decision: Rejected

Resuming graph execution...

--- [Node] Responding to Customer ---

-> Logged interaction to SQLite memory (Thread: demo_thread_001, Dept: Billing)

=== Final Response to Customer ===

Routed Department: Billing

Response: Your request for a refund has been reviewed by our supervisor and declined. Reason: High Risk

=====

Press Enter to proceed to the next query...
|

Query 5

```

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DEMONSTRATION QUERY 5/5
-> Applying Supervisor Decision: Rejected
Resuming graph execution...
--- [Node] Responding to Customer ---
-> Logged interaction to SQLite memory (Thread: demo_thread_001, Dept: Billing)

=== Final Response to Customer ===
Routed Department: Billing
Response: Your request for a refund has been reviewed by our supervisor and declined. Reason: High Risk
=====

Press Enter to proceed to the next query...

-----
DEMONSTRATION QUERY 5/5

=== Final Response to Customer ===
Routed Department: Billing
Response: Your request for a refund has been reviewed by our supervisor and declined. Reason: High Risk
=====

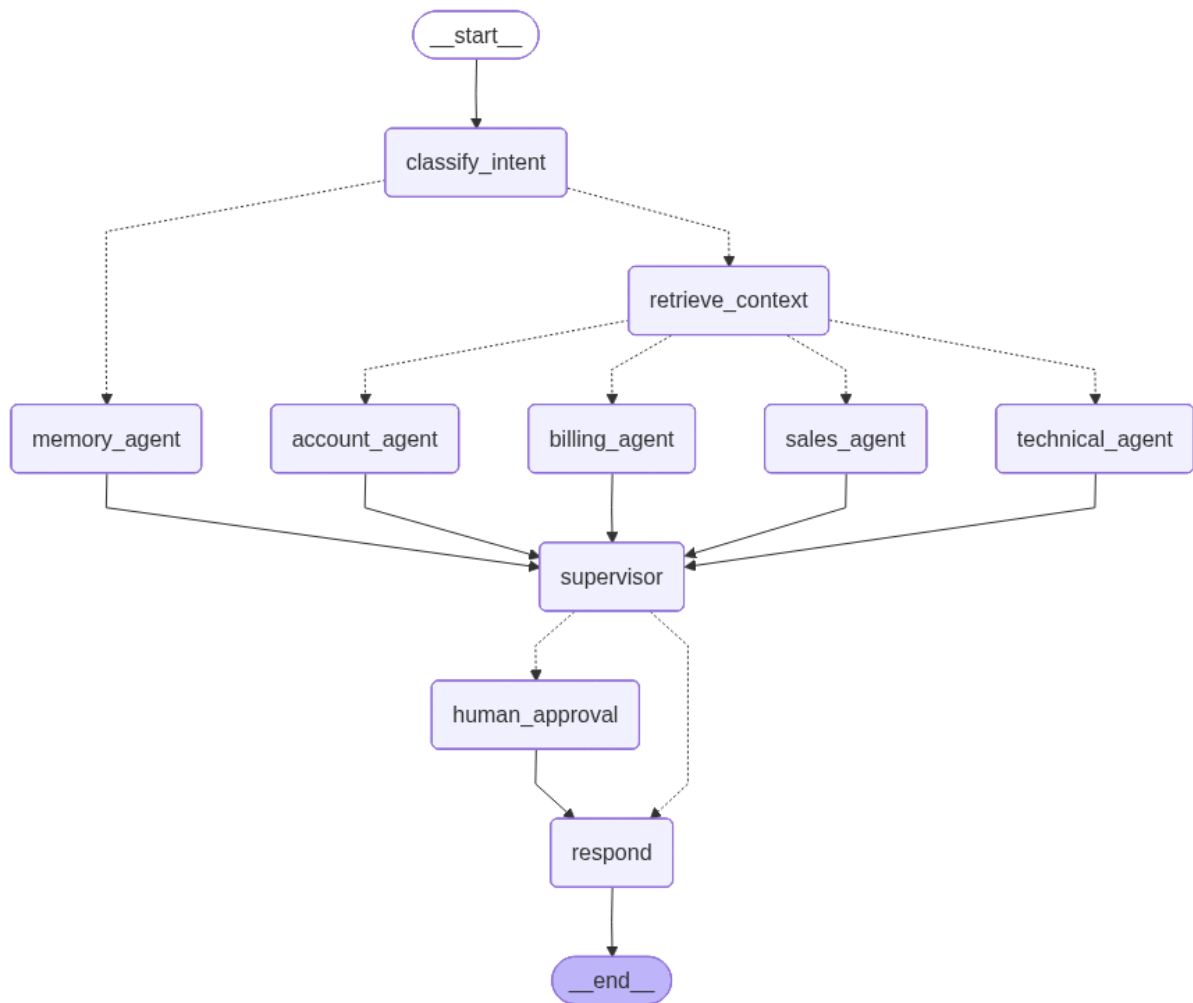
Press Enter to proceed to the next query...

-----
DEMONSTRATION QUERY 5/5
=== Final Response to Customer ===
Routed Department: Billing
Response: Your request for a refund has been reviewed by our supervisor and declined. Reason: High Risk
=====

Press Enter to proceed to the next query...

```

Workflow Diagram



Database

memory.db								
Rows: 9								
	id	thread_id	customer...	query	departm...	response	timestamp	
1	1	demo_thread_001	David	What are the pricing plans available for your software?	Sales	We offer three subscription plans tailored to different b...	2026-06-28 13:...	
2	2	demo_thread_001	David	I forgot my account password.	Account	To reset your password, please follow these steps: 1. Op...	2026-06-28 13:...	
3	3	demo_thread_001	David	My application crashes whenever I upload a file.	Technical	I'd be happy to help you troubleshoot the issue with yo...	2026-06-28 13:...	
4	4	demo_thread_001	David	I need a refund for my annual subscription.	Billing	To assist you with your refund request for your annual s...	2026-06-28 13:...	
5	5	demo_thread_001	David	What was my previous support issue?	Memory	Here is the polished response: Interaction #1: - Custome...	2026-06-28 13:...	
6	6	demo_thread_001	David	What are the pricing plans available for your software?	Sales	We offer three subscription plans tailored to different b...	2026-06-28 13:...	
7	7	demo_thread_001	David	I forgot my account password.	Account	To reset your password, please follow these steps: 1. Op...	2026-06-28 13:...	
8	8	demo_thread_001	David	My application crashes whenever I upload a file.	Technical	I'd be happy to help you troubleshoot the issue with yo...	2026-06-28 13:...	
9	9	demo_thread_001	David	I need a refund for my annual subscription.	Billing	Your request for a refund has been reviewed by our sup...	2026-06-28 13:...	
+	10							

```

=== SQLite database: customer_interactions ===
| ID | Thread ID | Customer | Dept | Query |
-----|-----|-----|-----|-----|
| 1 | demo_thread_001 | David | Sales | What are the pricing plans ... |
| 2 | demo_thread_001 | David | Account | I forgot my account password. |
| 3 | demo_thread_001 | David | Technical | My application crashes when... |
| 4 | demo_thread_001 | David | Billing | I need a refund for my annu... |
| 5 | demo_thread_001 | David | Memory | What was my previous suppor... |
| 6 | demo_thread_001 | David | Sales | What are the pricing plans ... |
| 7 | demo_thread_001 | David | Account | I forgot my account password. |
| 8 | demo_thread_001 | David | Technical | My application crashes when... |
| 9 | demo_thread_001 | David | Billing | I need a refund for my annu... |
| 10 | demo_thread_001 | David | Sales | What are the pricing plans ... |
| 11 | demo_thread_001 | David | Account | I forgot my account password. |
| 12 | demo_thread_001 | David | Technical | My application crashes when... |
| 13 | demo_thread_001 | David | Billing | I need a refund for my annu... |

Press Enter to return to main menu...

```

SQLite Schema

Custom Human-Readable Interaction Logs

```

CREATE TABLE customer_interactions (
  id INTEGER PRIMARY KEY AUTOINCREMENT,
  thread_id TEXT NOT NULL,
  customer_name TEXT,
  query TEXT NOT NULL,
  department TEXT,
  response TEXT NOT NULL,
  timestamp DATETIME DEFAULT CURRENT_TIMESTAMP
);

```

2. LangGraph Persistence Checkpoint Tables

These tables are automatically created and managed by `SqliteSaver` from the LangGraph library to track graph states, variables, and checkpoint history:

- `checkpoints`: Stores the serialized checkpoint data for the state at each step.
- `writes`: Tracks intermediate writes made by nodes during graph executions.
- `blobs`: Stores large objects/serialized states associated with checkpoints.

EXIT

Select an option:

- [1] Run Automated Demonstration (5 Sample Queries)
- [2] Start Interactive Chat Session
- [3] View SQLite Database Log Table
- [4] Exit

Enter choice (1-4): 4

Thank you for using the support system. Goodbye!

(venv) E:\AI-Powered Customer Support Automation System>